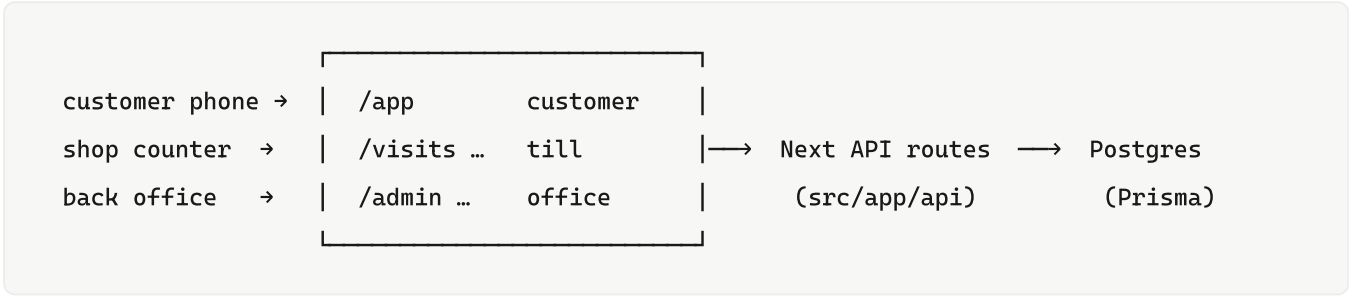
# Design

How the system is put together, and why. Written to be read by whoever picks this up next — including future me.

## Shape

One Next.js application, one Postgres database, three faces:




Everything shares one database and one set of rules. The rules themselves are plain functions in `src/lib`, with unit tests, and the API routes are thin wrappers over them. That is deliberate: a rule that lives in a screen gets copied and then diverges.

| Module | Holds |
|---|---|
| `lib/booking.ts` | Free-slot maths, opening hours, shop-local time. Pure. |
| `lib/booking-server.ts` | Booking settings, hours, and finding free times in the database |
| `lib/booking-create.ts` | Saving an appointment — the one path both staff and customers use |
| `lib/payments.ts` | How a visit was paid: cash, card, InstaPay, or a mix. Pure. |
| `lib/finance.ts` | The money model: commission, expenses and advances by account |
| `lib/session-policy.ts` | How long a login lasts, per role. Pure. |
| `lib/roles.ts` | Who is staff, which paths belong to which door. Pure. |

## Roles

| Role | Signs in with | Sees |
|---|---|---|
| `OWNER` | email | everything |
| `CASHIER` | email | the till, the diary, expenses, the queue |
| `BARBER` | email | their own day, their own customers, their own earnings |
| `CUSTOMER` | phone number | their own bookings and profile, nothing else |

Three doors, enforced in `middleware.ts`: shop paths need a staff role, `/app` needs a customer, `/barber` needs a barber. The API enforces the same thing independently (`requireStaff`, `requireBarber`, `requireCustomer`), because middleware protects pages, not data.

A barber may mark someone arrived, started or a no-show, and nothing else: taking the money stays at the till, and they cannot move, cancel or re-price a booking. That is what keeps the diary and the takings from drifting apart.

> A path check on the prefix `"/app"` once let a customer open `/appointments`, the shop's own screen. Paths are matched segment by segment now (`isCustomerPath`), with tests — and the same care applies to `/barber` against the shop's `/barbers` screen.

## The data model

The money side already existed; booking was added beside it.

```
Customer ──< Appointment >── Barber
                 │
                 ├──< AppointmentService   (name, price, length as quoted)
                 │
                 └─── Visit                (set when it is rung up)
                          │
                          ├──< VisitItem
                          └─── CashDrawerSession
```

**An appointment holds a chair. A visit is money.** They meet at `Appointment.visitId`, which is set once, when the customer pays. Until then the appointment carries only a quote. Ringing up is one

transaction: the visit is created and the appointment is stamped with it, so an appointment can never become two visits, and the shop can change the price at the till without rewriting what was quoted.

A customer who is served and leaves without paying is closed as `COMPLETED_UNPAID`: the chair was used, so the time stays taken, but no visit exists and no money is counted. If they come back and settle, it can still be rung up.

**A walk-in is the same appointment**, with `source = WALK_IN`, starting now, possibly with nobody assigned. That is what keeps one diary, one set of takings and one customer history however the customer arrived.

Decisions worth keeping:

- **Lines are snapshots.** `AppointmentService` copies the service name, price and length at booking time, as `VisitItem` already did. Raising the price of a haircut never rewrites last week's booking.
- **Cancelled frees the chair; completed does not.** One list of statuses (`BLOCKING_STATUSES`) decides this, used by the slot maths and the clash check alike.
- **"Any barber" is resolved at booking time**, not left vague: the customer gets a named barber immediately, so the shop can see who is doing what.
- **Hours are layered.** `ShopHours` is the week; `BarberWorkingHours` narrows it for one barber, never widens it. A barber with no rows simply works the shop's hours, so nothing has to be set up before booking works.
- **A barber with no `BarberService` rows does every service.** Same reason.

## Time

The shop thinks in local time — "Saturday, 11:00 to 23:00" — and Egypt changes its clocks twice a year. So opening hours are stored as minutes from midnight with a weekday, and turned into real moments through the shop's timezone (`Africa/Cairo`, a setting). `shopTime()` resolves the offset twice so it stays right on the changeover day; the tests pin both a summer and a winter date.

Everything stored in the database is a UTC instant. Only the boundaries convert.

## Free times

`availableSlots()` walks the working window on the shop's grid (every 15 minutes by default) and keeps a start time only if the whole service fits before closing without touching anything busy. Busy means appointments that still hold the chair, plus time off, each widened by the buffer.

The same function answers the customer app, the shop's booking modal and the reschedule screen. There is no second implementation to disagree with it.

## Double booking

Free times can be stale by the time someone taps one, so the check that matters happens on save, inside a `Serializable` transaction: the clash query and the insert are one atomic step. The loser of a race gets 409 "That time has just been taken", which the screens show as a message rather than an error.

## Offers

An offer is a notice, not a rule. It is shown in the customer app while it is running, and it never touches a price: the till still charges whatever the shop decides at the counter. That keeps the property everything else rests on — the takings are a record of what happened, not of what was advertised.

## Money

Unchanged by the booking work, and worth restating because everything eventually lands here:

- Revenue is visit totals; tips belong to the barber and are tracked apart.
- Three accounts: **cash** (the drawer), **card** (Visa, less its commission), **InstaPay** (a bank transfer, no commission, never in the drawer). Expenses and advances come out of whichever account paid them — `outflowAccount()` decides, in one place.
- Advances are payroll, not expenses: they move money without reducing revenue.
- A tip that was not handed over in cash is paid to the barber out of the drawer, and recorded as a movement.

## Reminders

A booking is worth nothing if the customer forgets it, so the app reminds them on the phone itself — a web push notification, which costs nothing per message and needs no provider, sender name or contract.

- A customer turns it on once, from their profile. The phone hands us a subscription (an address at Apple's, Google's or Mozilla's push service); we keep it against that customer, one row per phone.
- On an iPhone, notifications only exist once the app has been added to the Home Screen. The switch says so in as many words rather than failing silently.
- Two messages: **booking confirmed**, sent the moment it is booked, whoever booked it, and **a reminder**, sent the evening before. Both are written by pure functions in `reminders.ts`, in English or Arabic, in shop time.

- A nightly run ( `/api/cron/reminders` , on the scheduler, behind `CRON_SECRET` ) picks up everything starting within the next 20 hours that has not been reminded, sends, and stamps `reminderSentAt` . The stamp goes on even when the customer has no phone subscribed, so nobody is retried for ever.
- Sending never blocks a booking, and never fails one: it is fire-and-forget, and a subscription the push service reports as gone is deleted rather than retried nightly.

## Reviews

Stars, and a line if they feel like writing one. The rules are deliberately narrow, because a rating nobody believes is worse than no rating at all:

- Only a finished appointment can carry a review — including one they left without paying, since they still sat in the chair. Cancelled and missed appointments are silent.
- One per appointment. Writing it again is a change of mind, not a second opinion: the row is updated.
- The window closes after a month. Nobody is asked in March what they thought of a haircut in January.
- The evening run asks yesterday's customers once, on the phone. Ignoring the ask is an answer too, so it is never repeated.

The shop's only power over a review is `hiddenAt` : the owner can take an abusive one off the app, where it counts for nothing and is shown to nobody, but it is not deleted and the activity log records who hid it. A shop that can quietly delete criticism has a rating worth nothing, including to itself.

Averages are `null` when nobody has said anything — never zero, which reads as "terrible" rather than "new".

## Loyalty

The paper punch card without the paper. A stamp for every visit the customer paid for; after however many the shop sets, a reward is waiting.

Two things it deliberately does not do:

- It never changes a price. A full card is a note on the till screen while the customer stands there; the cashier decides what to charge and ticks "reward given", which records it. The takings stay a record of what was taken, which is the same reason offers are notices rather than rules.
- It never counts anything but a paid visit — a `Visit` row. A booking, a no-show, or someone who left without paying earns nothing.

Rewards earned are `floor(visits / required)`; rewards given are rows in `LoyaltyRedemption`; what is waiting is the difference, never below zero — a reward handed over early is the shop's kindness, not the customer's debt.

## Insights

A second reading of the same rows the money is made of: when the shop is busy, who did the work, how many faces were new, and whether people come back. Nothing here has its own store of numbers, so an insight can never disagree with the day journal.

Two of the questions cannot be answered from a window, and are not:

- **New versus returning** compares each visit against that customer's first visit ever. Counted inside the window instead, everybody would look new the moment the window opened.
- **Retention** reads each customer's whole history: how many came back at all, the typical gap between visits, and which regulars have not been in for two months — the last of which is the shop's best call list.

Times are bucketed in shop time, so a visit at half past midnight belongs to the night the shop worked.

## Customer groups

Six groups, worked out from the visits every time they are asked for: newcomers, regulars, slipping away, lapsed, big spenders, and the ones who tried the shop once. Nothing is stored — a customer joins a group by living their life and leaves it the same way, so a group can never go stale behind the shop's back.

The thresholds are written down in one place (`segments.ts`) because a shop that cannot say exactly what "a regular" means will argue with the number on the screen. Slipping and lapsed are kept apart deliberately, so the same person is not chased twice.

The owner can send one group a notification. Three limits, because the quickest way to lose someone's notifications for ever is to send them something they did not ask for:

- the owner alone, never the cashier and certainly never a customer;
- the same group at most once every twelve hours;
- only customers who turned notifications on — and the screen says how many that is before the message is written, which is usually far fewer than the group.

Every send is written to the activity log with the words that were sent. For everyone else, the screen copies the phone numbers so the shop can ring them.

## Sessions

Shop devices sit on a counter, so a staff login ends after 2 hours, or 30 minutes idle. A customer's phone is their own, so a customer login lasts as long as they keep using it (30 days idle). One policy module, `role` as the input; the token's own expiry carries it, so the middleware enforces it too.

## Conventions

- Screens are client components; the API is the security boundary.
- Every route checks its own permission. None trusts the middleware.
- Anything that rewrites or removes money writes an `AuditLog` line, fire-and-forget.
- Both languages, always: a new string is added to `messages/en.ts` and `messages/ar.ts` together, and the type makes a missing Arabic key a compile error.
- New rules arrive as a tested function in `lib/` first, then a screen.

---